

A CRT Carrier Bound for an LCM Recurrence: Exact Verification Through $10^{12} + 2$

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Abstract

Define $x(1) = 1$ and $x(n) = 2x(n-1) + \text{lcm}(x(n-1), n)$ for $n \geq 2$, and put $a(n) = x(n+1)/x(n) - 2$. A conjecture recorded in OEIS A135508 says that $a(p-1) = p$ whenever p is prime and $p-2$ is not prime. We prove the finite, computer-assisted version for every prime $p \leq 1,000,000,000,002$. The main reduction is a carrier lemma: if $q \geq 5$ is prime and a prime $s = mq - 2 \geq 11$ satisfies $1 \leq m \leq q$ and $s \equiv 5 \pmod{6}$, then the first index $T(q)$ for which $q \mid x(T(q))$ exists and obeys $T(q) \leq s \leq q^2 - 2$. An exhaustive certificate, checked by a complete Eratosthenes sieve, supplies such a carrier for all primes $q \leq 10^6$. The universal conjecture remains unresolved.

1 Statement and scope

For positive integers, let gcd and lcm denote the positive greatest common divisor and least common multiple. Consider

$$x(1) = 1, \quad x(n) = 2x(n-1) + \text{lcm}(x(n-1), n) \quad (n \geq 2), \quad (1)$$

and

$$a(n) = \frac{x(n+1)}{x(n)} - 2 \quad (n \geq 1). \quad (2)$$

The quotient in (2) is integral. Indeed, on setting

$$r_k = \frac{k}{\text{gcd}(x(k-1), k)}, \quad (3)$$

the recurrence becomes

$$x(k) = x(k-1)(r_k + 2), \quad a(k-1) = r_k. \quad (4)$$

In particular, $x(k-1) \mid x(k)$.

The motivating conjecture appears in OEIS A135508 [5] and in the empirical study of Schepke [4, Sec. 3.2.2]. Cloitre's recent analysis treats this exact $K = 2$ recurrence and its twin-prime structure, while leaving its prime-index conclusions conditional [1, Sec. 6]. The conjecture asserts

$$p \text{ prime and } p-2 \text{ not prime} \implies a(p-1) = p. \quad (5)$$

Here 0 and 1 are not prime, so the antecedent includes $p = 2, 3$. Our result is the following explicitly finite statement.

Theorem 1 (Finite recurrence theorem). *For every prime $p \leq 1,000,000,000,002$, if $p - 2$ is not prime, then $a(p - 1) = p$.*

The proof has a symbolic part and an exact finite certificate. It does not establish (5) for arbitrary primes.

2 A local acquisition criterion

For a prime q , define

$$T(q) = \min\{n \geq 1 : q \mid x(n)\}$$

whenever this set is nonempty. We first locate exactly how a prime can enter the recurrence immediately before its own index.

Proposition 2 (Local criterion). *If $p \geq 5$ is prime, then*

$$p \mid x(p - 1) \iff \gcd(x(p - 3), p - 2) = 1. \quad (6)$$

Consequently,

$$a(p - 1) = p \iff \gcd(x(p - 3), p - 2) > 1. \quad (7)$$

Proof. For $2 \leq k \leq p - 1$, equation (3) gives $r_k \mid k$ and

$$3 \leq r_k + 2 \leq k + 2 \leq p + 1 < 2p.$$

Since $x(1) = 1$, the product formula (4) shows that $p \mid x(p - 1)$ precisely when p divides one of the multipliers $r_k + 2$ in this range. Such a multiplier must equal p , so $r_k = p - 2$. Because $r_k \mid k$ and $2(p - 2) > p - 1$, this is possible only for $k = p - 2$. It occurs there exactly when $\gcd(x(p - 3), p - 2) = 1$, which proves (6).

For any prime p , equation (2) gives

$$a(p - 1) = \frac{p}{\gcd(x(p - 1), p)}.$$

Thus $a(p - 1) = p$ if and only if $p \nmid x(p - 1)$. Combining this dichotomy with (6) proves (7). $\square$

3 Forced CRT carriers

The next lemma turns a prime in one particular residue class into a factor of a later recurrence value.

Lemma 3 (Carrier lemma). *Let $q \geq 5$ be prime. Suppose that integers m, s satisfy*

$$1 \leq m \leq q, \quad s = mq - 2 \geq 11, \quad s \text{ prime}, \quad s \equiv 5 \pmod{6}.$$

Then $T(q)$ exists and

$$T(q) \leq s \leq q^2 - 2.$$

Proof. Direct calculation gives

$$x(2) = 4, \quad x(3) = 20, \quad x(4) = 60.$$

Divisibility in (4) therefore implies $3 \mid x(j)$ for every $j \geq 4$. Since $s \equiv 5 \pmod{6}$ and $s \geq 11$, we have $3 \mid s - 2$ and $s - 3 \geq 4$. Hence

$$\gcd(x(s - 3), s - 2) \geq 3.$$

Proposition 2, applied to the prime s , now gives $s \nmid x(s - 1)$. It follows that $\gcd(x(s - 1), s) = 1$, so $r_s = s$. The multiplier at step s is consequently

$$r_s + 2 = s + 2 = mq.$$

Thus $q \mid x(s)$, which proves $T(q) \leq s$. Finally, $m \leq q$ gives $s = mq - 2 \leq q^2 - 2$. $\square$

The congruences $s \equiv -2 \pmod{q}$ and $s \equiv 5 \pmod{6}$ specify a reduced residue class modulo $6q$. The second congruence is essential in this argument: it forces the common factor 3 that makes the step at s active.

4 Exact certificate and proof of the finite theorem

For each prime q with $5 \leq q \leq 10^6$, the certificate searches

$$m = q \bmod 6, \quad q \bmod 6 + 6, \quad q \bmod 6 + 12, \dots, \quad m \leq q, \tag{8}$$

and records the least m for which $s = mq - 2 \geq 11$ is prime. Because a prime $q \geq 5$ is congruent to 1 or 5 modulo 6, condition $m \equiv q \pmod{6}$ gives $s \equiv q^2 - 2 \equiv 5 \pmod{6}$.

The serialized table contains one row for each of the 78,496 primes $5 \leq q \leq 10^6$. Across all rows, 375,964 candidate values were tested. The largest recorded least multiplier is $m = 277$ at $q = 379,123$, and the largest witness prime is $s = 243,937,433$ at $q = 995,663$ with $m = 245$.

The generator uses a complete sieve for the q -domain and deterministic 64-bit Miller–Rabin screening for candidate values of s . The decisive verification is structurally separate: a C++ program constructs a complete odd Eratosthenes sieve through 243,937,433 and checks all of the following exactly:

- the table has exactly one ordered row for every prime $5 \leq q \leq 10^6$ and no additional rows;
- every displayed s is prime, satisfies $s = mq - 2 \geq 11$ and $s \equiv 5 \pmod{6}$, and has $m \leq q$;
- every earlier candidate in (8) with $s \geq 11$ is composite.

It checked the same 375,964 candidate values without any failure. The minimality check is stronger than the carrier lemma requires, but it makes the finite search boundary auditable.

Theorem 4 (Certified carrier range). *For every prime $q \leq 10^6$, $T(q)$ exists and*

$$T(q) \leq q^2 - 2.$$

Proof. For $5 \leq q \leq 10^6$, the exact certificate supplies the hypotheses of Lemma 3. The two omitted cases are direct: $T(2) = 2 \leq 2^2 - 2$ and $T(3) = 4 \leq 3^2 - 2$. $\square$

Proof of Theorem 1. The boundary cases hold directly: $x(2) = 4$ gives $a(1) = 2$, while $x(3) = 20$ gives $a(2) = 3$. Now let $p \geq 5$ be prime, $p \leq 10^{12} + 2$, and suppose that $n = p - 2$ is not prime. Then $n > 1$ is odd and hence composite. It has an odd prime divisor

$$q \leq \sqrt{n} \leq 10^6.$$

By Theorem 4,

$$T(q) \leq q^2 - 2 \leq n - 2 = p - 4 < p - 3.$$

Divisibility persists in (4), so $q \mid x(p-3)$. Since $q \mid p-2$ as well, $\gcd(x(p-3), p-2) > 1$. Proposition 2 then yields $a(p-1) = p$. $\square$

5 Context, limitations, and reproducibility

The OEIS entry tabulates the sequence only through index 10,000 [5]. Schepke’s thesis defines the same recurrence (under the notation b_2, q_2) and discusses the observed one-or-prime and twin-prime behavior [4, Sec. 3.2.2]. In the direct sources inspected for this project, neither source supplied the carrier lemma or the finite bound above. Ruiz–Cabello’s coefficient-1 LCM recurrence uses the neighboring class $-1 \pmod{q}$ and exposes the analogous quadratic least-prime barrier [3, Sec. 2]. For the exact coefficient-2 recurrence, Cloitre replaces that class by $-2 \pmod{q}$ and proves only conditional $K = 2$ density and prime-index statements [1, Secs. 6.5–6.6]; that source reports direct computation only through index 20,000. Thus these closest sources do not contain the unconditional carrier certificate or the finite theorem through $10^{12} + 2$ proved here. This is a bounded source comparison, not a claim of priority.

A universal proof by the same route would follow from showing, for every odd prime q , that the specified residue class modulo $6q$ contains a prime below q^2 . General unconditional Linnik-type estimates do not give this quadratic threshold; for example, the exponent 5.18 established by Xylouris [6] is too large. Sharper least-prime bounds are the subject of conjectural and computational work such as [2], but no all- q quadratic theorem used here is available from those sources. The finite certificate cannot replace that missing quantifier, so the original conjecture remains unresolved beyond the stated range.

The computational evidence consists of a Python generator, its serialized witness table, and a separate C++ verifier. From the project root, these three companion files can be regenerated and checked with

```
python3 evidence/crt_witness_scan.py
clang++ -std=c++20 -O2 -Wall -Wextra -pedantic \
    evidence/verify crt_witnesses.cpp -o verify crt_witnesses
./verify crt_witnesses evidence/crt_witnesses_q_le_1000000.csv audit.json
```

The verifier’s conclusion depends only on integer arithmetic and its complete sieve over the explicitly stated range.

Computational and AI assistance. Generative-AI assistance (OpenAI Codex) was used to explore the reduction, implement and test the Python and C++ programs, and draft the manuscript. A separate fresh model context rederived the symbolic argument and ran structurally different exact checks; this was a robustness check, not independent peer review. The author selected the stated scope, reviewed the mathematical and computational evidence, and takes responsibility for the result. A user-supplied unpublished item on AI chatbot code generation [7] is cited only in connection with the type of coding assistance used; it supplies no mathematical premise, prior-art claim, or evidence of novelty.

References

- [1] Benoit Cloitre. Primes in LCM recurrences. arXiv preprint arXiv:2510.18891v3, 2026. Version 3, revised 21 April 2026; <https://arxiv.org/abs/2510.18891v3>.
- [2] Andrew Fiori. The least prime in arithmetic an progression. arXiv preprint arXiv:2404.02329, 2024. <https://arxiv.org/abs/2404.02329>.
- [3] Serafín Ruiz-Cabello. On the use of the least common multiple to build a prime-generating recurrence. *International Journal of Number Theory*, 13(4):819–833, 2017. Preprint <https://arxiv.org/abs/1504.05041>.
- [4] Markus Schepke. Über primzahlerzeugende folgen. Bachelor’s thesis, Universität Hannover, 2009. Available at https://primes.recommend.games/schepke_primzahlerzeugende_folgen.pdf.
- [5] The OEIS Foundation. A135508. The On-Line Encyclopedia of Integer Sequences, 2026. Entry and table available at <https://oeis.org/A135508>; accessed 6 September 2026.
- [6] Triantafyllos Xylouris. On the least prime in an arithmetic progression and estimates for the zeros of Dirichlet L-functions. *Acta Arithmetica*, 150(1):65–91, 2011. <https://doi.org/10.4064/aa150-1-4>; preprint <https://arxiv.org/abs/0906.2749>.
- [7] Zijian Zeng. AI Chatbot for Code Generation. User-supplied unpublished manuscript; no journal, DOI, or abstract was supplied, 2025.